

Probe-EM: Targeted Neuron Tracing via Training-Free Semantic Verification

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CODE



VIDEO DEMO



BACKGROUND

Automated EM reconstructions fragment neurons, making manual proofreading a major bottleneck in large-scale connectomics. Supervised connectivity classifiers require annotated segment pairs and often scan non-target regions. Probe-EM reformulates correction as targeted, seed-driven retrieval: probe nearby fragments, verify connectivity, and grow the neuron of interest. It handles both intra-slice boundary splits and inter-slice gaps caused by misalignment, tissue loss or changing neurite morphology.

Contributions

Skeleton-guided exploration: Retrieves nearby fragments to trace a target neuron from a seed.

Dimension-aware verification: Resolves intra- and inter-slice splits without connectivity-specific training.

Interactive proofreading: Reduces annotation time by **33.4%** while improving reconstruction accuracy.

METHOD

HSS: targeted probing

Initialize a queue with the seed segment. Skeleton endpoints propose nearby fragments within a 500 nm radius. Endpoint counts and density flag topological anomalies associated with mergers or glial misconnections. Prune suspect branches and enqueue verified segments for further exploration.

PEC: intra-slice splits

Anchor points establish segment identity, probe points test boundary permeability, and rear points preserve mask integrity. Predict masks in both directions and test high/low overlap thresholds. Confirm connectivity through consensus across five randomized trials.

ASP: inter-slice splits

Treat z-slices as a short video. Memory-based propagation follows neurite morphology across sections. Propagate the source mask over $N = 5$ frames and measure candidate occupancy. Accept a connection when maximum coverage over the window exceeds 0.6.

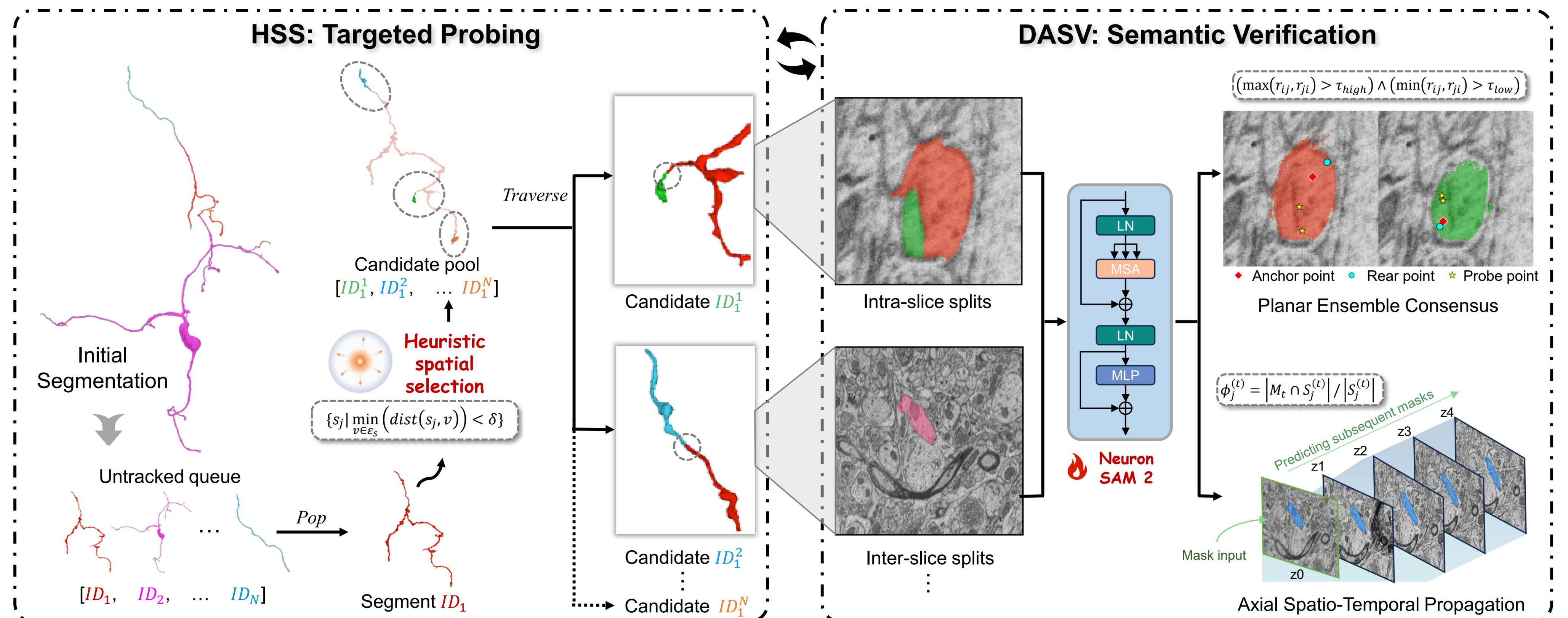


Fig. 1. Overview of Probe-EM. The framework facilitates iterative tracing of complete neuronal morphologies through a synergistic probing-verification cycle.

NeuroSAM 2 adapts SAM 2 [2] on EMNeuron [3], with no connectivity labels. All experiments use one RTX 4090.

RESULTS

Quantitative tracing on the mouse suprachiasmatic nucleus (SCN) [1]

Method	Axon fascicle tracing			Soma-seeded tracing		
	Recall	Precision	F1	Recall	Precision	F1
Voxel-based CNN	0.542	0.902	0.571	0.245	0.428	0.293
Geometric PointNet	0.509	0.485	0.288	0.181	0.498	0.210
Multimodal Fusion	0.421	0.650	0.436	0.160	0.372	0.202
Probe-EM (SAM 2)	0.736	0.494	0.467	0.492	0.312	0.310
Probe-EM (NeuroSAM 2)	0.753	0.626	0.595	0.705	0.565	0.544

Qualitative comparison

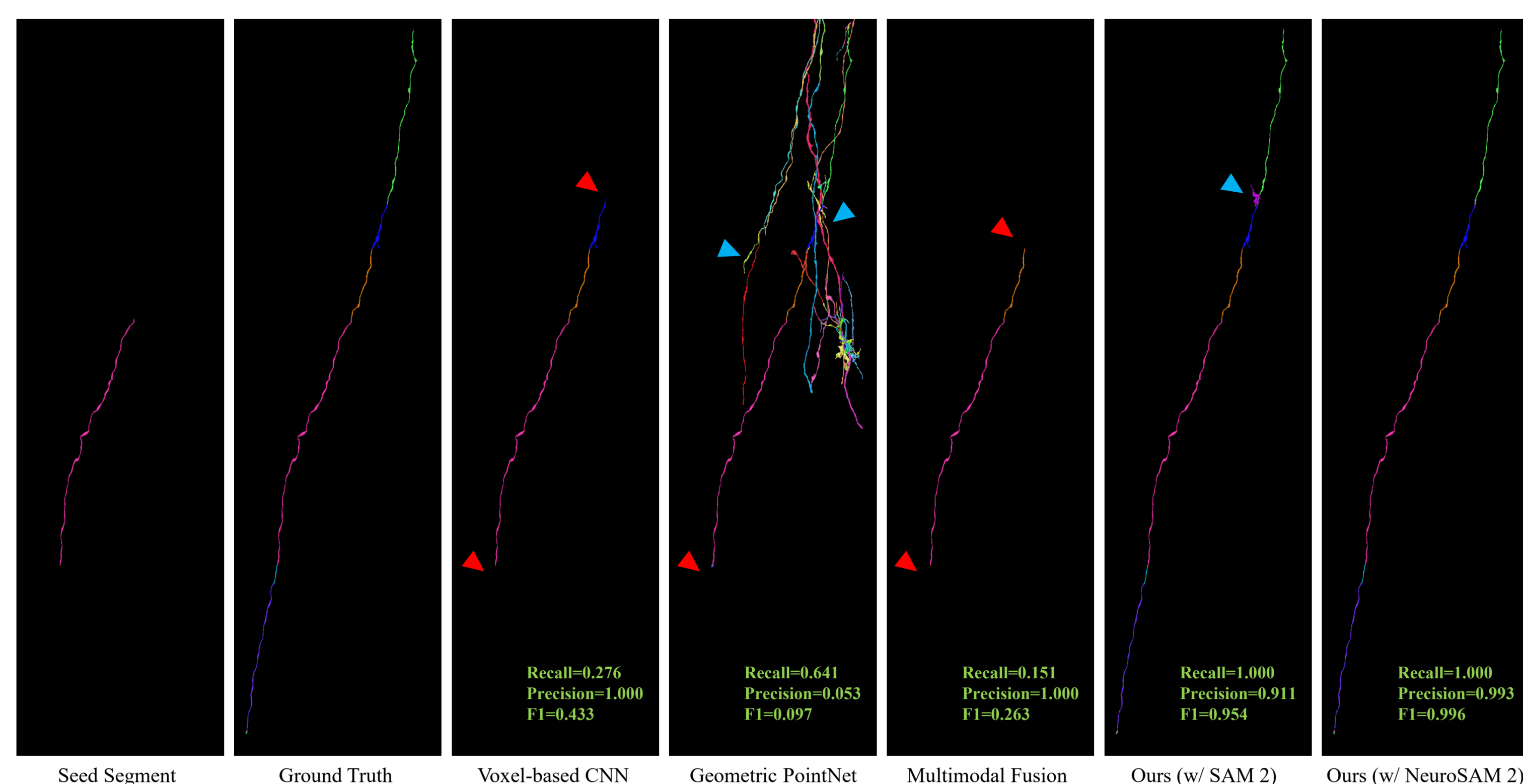


Fig. 2. Qualitative visualization of neuron tracing results.

Ablation study

Configuration	PEC	ASP	GPS	N	Recall	Precision	F1
w/o PEC		✓	✓	5	0.595	0.605	0.483
w/o ASP	✓		✓	5	0.020	0.800	0.036
w/o GPS	✓	✓		5	0.588	0.346	0.328
Ours (N = 3)	✓	✓	✓	3	0.611	0.624	0.484
Ours (Default)	✓	✓	✓	5	0.691	0.639	0.586
Ours (N = 8)	✓	✓	✓	8	0.589	0.396	0.371

Human-in-the-loop efficiency

Annotator	Manual Only		w/ Probe-EM		Improvement	
	Time (min)	F1 Score	Time (min)	F1 Score	Δ Time	Δ F1
Annotator 1	52.6	0.874	29.2	0.892	-44.5%	+2.1%
Annotator 2	31.6	0.827	20.0	0.974	-36.7%	+17.8%
Annotator 3	30.2	0.894	24.5	0.899	-18.9%	+0.6%
Average	38.1	0.865	24.6	0.922	-33.4%	+6.8%

CONCLUSION

Probe-EM combines targeted spatial search with semantic verification to recover neuronal morphology without connectivity-specific training. It achieves the highest F1 in both tracing tasks and reduces manual proofreading time by 33.4% while improving mean F1 from 0.865 to 0.922.

References

- [1] Liu et al. Connectomic Organization of the Suprachiasmatic Nucleus. bioRxiv (2024).
- [2] Ravi et al. SAM 2: Segment Anything in Images and Videos. arXiv:2408.00714 (2024).
- [3] Zhang et al. SegNeuron: 3D Neuron Instance Segmentation in Any EM Volume with a Generalist Model. MICCAI (2024).